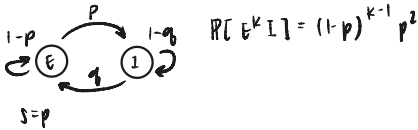


Markov Models and HMMs:



Maximum Likelihood Estimates:

$\hat{\theta}_{MLE} = \arg\max_{\theta} L(\theta) = \arg\max_{\theta} \log L(\theta)$

(1) setup: $L(\theta) = P[\text{data} | \theta]$
if data i.i.d. $P[\text{data} | \theta] = \prod_{i=1}^n P[X_i | \theta]$

(2) Find log likelihood:
 $\arg\max_{\theta} L(\theta) = \arg\max_{\theta} \log L(\theta)$

if i.i.d. $\log L(\theta) = \sum_{i=1}^n \log P[X_i | \theta]$

(3) Take derivative/gradient w.r.t. parameter being maximized + set to zero

(4) solve for parameters

Expectation Maximization:

↳ finds statistically overrepresented sequence motif based on assumption that there is one binding site in the sequence before each gene.

• Multiple Expectation Maximization for motif Elicitation (MEME): tool to discover motifs in a group of related DNA or protein sequences.

- (1) initialize probb matrix p
- (2) calculate expected vals for z (loc matrix)
- (3) use z to find better vals for p (max)
- (4) Iterate (2)+(3) until $\Delta p \leq \epsilon$

Position Weight Matrices:

↳ relies on each pos in motif being i.i.d. $\Rightarrow$ multiplying probs

pos	1	2	3	
A	0.2	0.4	0.2	$P[\text{CAT} \text{binding site}] = 0.4^3$ $P[\text{CAT}] = \frac{P[\text{CAT} \text{bs}] \times P[\text{bs}]}{P[\text{CAT}]}$ $P[\text{bs} \text{CAT}] = \frac{P[\text{CAT} \text{bs}] \times P[\text{bs}]}{P[\text{CAT} \text{bs}] \times P[\text{bs}] + P[\text{CAT} \text{bq}] \times P[\text{bq}]}$ $\text{bq} = 0.25$ $P[\text{bs}] = 0.01$ $P[\text{bq}] = 1 - 0.01 = 0.99$
C	0.4	0.2	0.2	
G	0.2	0.2	0.2	
T	0.2	0.2	0.4	

Find MLE of p:

$\hat{p}_{MLE} = \arg\max_p L(p; \text{data}) = \arg\max_p (1-p)^{k-1} p^2$

$\frac{d}{dp} (1-p)^{k-1} p^2 = -(k-1)(1-p)^{k-2} p^2 + (1-p)^{k-1} 2p = 0$

$2p(1-p)^{k-1} = (k-1)(1-p)^{k-2} p^2$

$2p(1-p) = (k-1)p^2 \Rightarrow p = \frac{2}{k+1}$

m i.i.d. RV $X_1, \dots, X_m \sim \text{Binom}(n, p)$. Find MLE for p:
 $L(p) = \prod_{i=1}^m \binom{n}{X_i} p^{X_i} (1-p)^{n-X_i} \rightarrow \log L(p) = \log \prod_{i=1}^m \binom{n}{X_i} p^{X_i} (1-p)^{n-X_i}$

$\log L(p) = \sum_{i=1}^m \log \binom{n}{X_i} + \sum_{i=1}^m X_i \log p + \sum_{i=1}^m (n-X_i) \log (1-p)$

$\hat{p}_{MLE} = \arg\max_p \log L(p) \rightarrow \frac{d}{dp} \log L(p) = \frac{\sum_{i=1}^m X_i}{p} - \frac{\sum_{i=1}^m (n-X_i)}{1-p}$

$(1-p) \sum_{i=1}^m X_i = p \sum_{i=1}^m (n-X_i)$

$\sum_{i=1}^m X_i - p \sum_{i=1}^m X_i = mnp - p \sum_{i=1}^m X_i$

$\sum_{i=1}^m X_i = mnp$

$p = \frac{\sum_{i=1}^m X_i}{mn}$

Probability:

- Binomial(n, p): $P[X=k] = \binom{n}{k} p^k (1-p)^{n-k}$, $E[X] = np$, $\text{Var}(X) = np(1-p)$
- Poisson(λ): $P[X=k] = \frac{\lambda^k}{k!} e^{-\lambda}$, $E[X] = \lambda$, $\text{Var}(X) = \lambda$
- disjoint = mutually exclusive $\Rightarrow$ not i.i.d.

col zero $\rightarrow$ bg probs

normalizing rows if z: $z_i = z_i / \sum (z_i \forall i)$

seq: CTG AATAGA $z = [0.055, 0, 0, 0.047, 0.002, 0.895, 0, 0, 0]$

pos	1	2	3	4
A	0.049	0.942	0.002	0.997

background: A: $\frac{4}{9}$ C: $\frac{1}{9}$ G: $\frac{2}{9}$ T: $\frac{2}{9}$

MISC:

most entropy = most random

Linear Algebra:

- $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times p}$: $\begin{bmatrix} I_n & 0 \\ 0 & AB \end{bmatrix} \rightarrow \begin{bmatrix} B & I_n \\ 0 & A \end{bmatrix}$
- diagonalizable iff n l.i.d. eigenvectors
- SVD: $A_{m \times n} = U_{m \times m} \Sigma_{m \times n} V_{n \times n}^T$
- (1) solve $\text{char}_{A^T A}(\lambda)$ $\hookrightarrow$ works for any matrix
- (2) Σ : $\sigma = \sqrt{\lambda}$ in dec order
- (3) V : corr EV/ONB $\vec{v}_1, \dots, \vec{v}_n$
- (4) U : $\vec{u}_i = \frac{1}{\sigma_i} (A^T A) \vec{v}_i$